Coding for Bayesian hierarchical model used in:

Elephants' fear of the human "super predator" transforms the African savanna and saves crops

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Bayesian hierarchical model

We modeled the probability of occurrence (ψ_{jt}) that elephants used camera site j (j = $1, 2, \ldots, 24$)during treatment period t ($= 1, 2$) as:

$$logit(f(\psi_{jt})) = \beta_0 + \beta_1 * S_j + \beta_2 * T_{jk} + \beta_3 * S_j * T_{jk} + \phi_\psi \quad (1)$$

where T_{jk} denotes the playback treatment (0 = control, 1 = human), S_j demoted grid site (0 = South, 1 = North), $S_j * T_{jk}$ denotes the interaction between grid site and playback treatment, $\boldsymbol{\beta}$ is a vector of parameter estimates, and ϕ_ψ captures the latent spatial structure of site use. True site use by elephants at camera site j during treatment period t was estimated as a latent binary variable:

$$z_{jt} \sim Bernoulli(\psi_{jt}) \quad (2)$$

For each camera site j and survey period k (k = 1, 2, …, 10 weeks), we modelled the number of independent elephant detections (y_{jk}) given site use as a Poisson-gamma mixture model representing a negative binomial distribution to relax the Poisson distribution assumption of equal dispersion through a single-parameter gamma-distributed random variable ρ_{jk}:

$$y_{jk} \sim Poisson(z_{jt} * \mu_{jk}) \quad (3)$$

$$\mu_{jk} = \rho_{jk} * \lambda_{jk} \quad (4)$$

$$\rho_{jk} \sim Gamma(\theta, \theta) \quad (5)$$

$$log(g(\lambda_{jk})) = \alpha_0 + \alpha_1 * S_j + \alpha_2 * T_{jk} + \alpha_3 * S_j * T_{jk} + \phi_\lambda \quad (6)$$

where λ_{jk} denotes the expected elephant detection frequency given site use, $\boldsymbol{\alpha}$ is a vector of parameter estimates, and ϕ_λ captures the latent spatial structure of detection frequency. Spatial random effects were modeled as multivariate normal processes where covariance matrices followed Gaussian spatial decay functions of pairwise distances between camera sites ($D_{jj'}$):

$$\phi_\psi \sim MVN(0, \Sigma_\psi, \rho_\psi) \quad (7)$$

$$h(\phi_\psi) = \Sigma_\psi^2 * exp\left[\frac{-D_{jj'}}{2 * \rho_\psi^2}\right] \quad (8)$$

$$\phi_\lambda \sim MVN(0, \Sigma_\lambda, \rho_\lambda) \quad (9)$$

$$l(\phi_\lambda) = \Sigma_\lambda^2 * exp\left[\frac{-D_{jj'}}{2 * \rho_\lambda^2}\right] \quad (10)$$

where ρ_ψ and ρ_λ denote the spatial range parameters controlling the rate of decay in covariance between camera traps and Σ_ψ and Σ_λ denote the marginal variance of spatial random effects. Parameter estimates for both submodels ($\boldsymbol{\alpha}, \boldsymbol{\beta}$) were drawn from diffuse normal priors (mean = 0, precision = 0.001), spatial parameters ($\Sigma_\psi, \rho_\psi, \Sigma_\lambda, \rho_\lambda$) were drawn from uniform

priors (0.01, 3), and the overdispersion parameter (ρ) was drawn from a log-uniform prior (-5,5). We ran three Markov Chain Monte Carlo chains of 300,000 iterations, with a burn-in of 250,000 and a thinning interval of 50, yielding 1,000 posterior samples per chain. Model convergences was verified via visual inspection of trace plots and the Gelman-Rubin statistic ($\hat{R} < 1.1$). Model fit was evaluated using posterior predictive checks based on Freeman-Tukey (p = 0.399) and chi-squared (p = 0.393) discrepancy statistics by comparing observed and simulated data via Bayesian p-values. Key derived quantities included mean detection frequencies (λ) and occurrence probabilities (ψ) for human and control treatments, mean value of rho (ρ), and the mean effective spatial autocorrelation of occurrence probability (ϕ_ψ) and intensity of site use (ϕ_λ).

Zero-Inflated Negative Binomial Model Random Variables:

y_{jk} – Observed detections of elephants at camera site j during survey period k

$y_{jk} \sim \text{Poisson}(z_{jt} * \rho_{jk} * \lambda_{jk})$

z_{jt} – Use of camera site j by elephants during treatment period t

$z_{jt} \sim \text{Bernoulli}(\psi_{jt})$

ψ_{jt} – Probability of elephants using camera site j during treatment t

(function of a deterministic model, see equation 1 above)

ρ_{jk} – Count overdispersion parameter

$\rho_{jk} \sim \text{Gamma}(\theta, \theta)$

λ_{jk} – Average number of detections by elephants at camera site j during survey period k

(function of a deterministic model, see equation 6 above)

– Vector of parameter estimates for covariates on λ

$\boldsymbol{\alpha} \sim \text{Normal}(\boldsymbol{\mu}_\alpha, \boldsymbol{\sigma}_\alpha)$

– Vector of parameter estimates for covariates on z

$\boldsymbol{\beta} \sim \text{Normal}(\boldsymbol{\mu}_\beta, \boldsymbol{\sigma}_\beta)$

ϕ_ψ – Latent spatial random effect for ψ_{jt}

$\varphi_\psi \sim \text{MVN}(0, \Sigma_\psi, \rho_\psi)$

ϕ_λ – Latent spatial random effect for λ_{jk}

$\varphi_\psi \sim \text{MVN}(0, \Sigma_\lambda, \rho_\lambda)$

θ – Gamma parameter on ρ_{ijk}

$\theta \sim \text{Uniform}(-5,5)$

$\boldsymbol{\mu}_\alpha$ – Vector of means of the distributions from which $\boldsymbol{\alpha}$ are drawn

$\sim \text{Normal}(0, 1000)$

σ_α – Vector of standard deviations of the distributions from which α are drawn
σ ~ Uniform(0, 100)

μ_β – Vector of means of the distributions from which β are drawn
μ_β ~ Normal(0, 1000)

σ_β – Vector of standard deviations of the distributions from which β are drawn
σ_β ~ Uniform(0, 100)

ρ_ψ – spatial range parameter on ϕ_ψ
ρ_ψ ~ Uniform(0.1, 3)

Σ_ψ – Marginal variance of the distribution from which ϕ_ψ is drawn
Σ_ψ ~ Uniform(0.01, 3)

ρ_λ – spatial range parameter on ϕ_λ
ρ_λ ~ Uniform(0.1, 3)

Σ_λ – Marginal variance of the distribution from which ϕ_λ is drawn
Σ_λ ~ Uniform(0.01, 3)

Zero-Inflated Negative Binomial Model JAGS Code:

```
model{

    # Prior for gamma distribution (rho)
    theta <- exp(logtheta)
    logtheta ~ dunif(-5,5)

    ############
    # Spatial Parameters for Gaussian Process
    # Range parameters
    rho.psi ~ dunif(0.1, 3)
    rho.lambda ~ dunif(0.1, 3)

    # Marginal variance parameters
    sigma2.psi ~ dunif(0.01, 3)
    sigma2.lambda ~ dunif(0.01, 3)

    ############
    # Priors for probability of occurrence parameters (psi)
    a0 ~ dnorm(0, 0.001)      #intercept
    a1 ~ dnorm(0, 0.001)      #site effect
    a2 ~ dnorm(0, 0.001)      #treatment effect
    a3 ~ dnorm(0, 0.001)      #site x treatment interaction

    ############
    # Priors for intensity of site use parameters (lambda)
    b0 ~ dnorm(0, 0.001)      #intercept
    b1 ~ dnorm(0, 0.001)      #site effect
```

```
b2 ~ dnorm(0, 0.001)      #treatment effect
b3 ~ dnorm(0, 0.001)      #site x treatment interaction

############
# Covariance matrices for spatial effects
for(j in 1:M){
    for(k in 1:M){
        Sigma.psi[j,k] <- sigma2.psi *
                          exp(-pow(D[j,k],2)/(2*pow(rho.psi,2))) +
                          1.0E-6 * equals(j,k)
        Sigma.lambda[j,k] <- sigma2.lambda *
                             exp(-pow(D[j,k],2)/(2*pow(rho.lambda,2)))+
                             1.0E-6 * equals(j,k)
    }
}
Omega.psi[1:M,1:M] <- inverse(Sigma.psi[,])
Omega.lambda[1:M,1:M] <- inverse(Sigma.lambda[,])

############
# Spatial random effects
phi.psi[1:M] ~ dmnorm(zeros[], Omega.psi[,])
phi.lambda[1:M] ~ dmnorm(zeros[], Omega.lambda[,])

# Vector of zeros for MVN mean
for(i in 1:M){
    zeros[i] <- 0
}

############
# Couples each k survey period to the appropriate treatment period
t.track<-c(1,1,1,1,1,2,2,2,2,2)

############
# Logistic regression submodel for probability of occurence (psi)
for(j in 1:M){                #loop through camera traps (1 to 24)
    for(t in 1:2){            #loop through treatment periods (1 to 2)
        z[j,t] ~ dbern(psi[j,t])

        logit(psi[j,t]) <- b0 + b1*Trt[j,t] + b2*S.psi[j,t] +
        b3*Trt[j,t]*S.psi[j,t] + phi.psi[j]

        loglik.psi[j,t] <- z[j,t] * log(psi[j,t] + 1.0E-10) +
                           (1 - z[j,t]) * log(1 - psi[j,t] + 1.0E-10)
    }
}

############
# Negative binomial submodel for intensity of site use (lambda)
for(j in 1:M){                  #loop through camera traps (1 to 24)
    for(k in 1:K){              #loop through survey periods (1 to 10)
        log(lambda[j,k]) <- a0 + a1*Trt[j,k] + a2*S[j,k] +
        a3*Trt[j,k]*S[j,k] + phi.lambda[j]

        rho[j,k] ~ dgamma(theta, theta)
```

```
        mu[j,k] <- rho[j,k]*lambda[j,k]

        y[j,k] ~ dpois(z[j,t.track[k]] * mu[j,k])

        loglik.lambda[j,k] <- logdensity.pois(y[j,k],
        z[j,t.track[k]] * mu[j,k])

        # Create new data for calculating Bayesian p-values
        y_new[j,k] ~ dpois(z[j,t.track[k]] * mu[j,k])

        # Calculate statistics for Bayesian p-values
        eval[j,k] <- z[j,t.track[k]] * mu[j,k]

        # Freeman-Tukey Residual
        Terr[j,k] <- pow(pow(y[j,k],.5) - pow(eval[j,k],.5),2)
        Terrnew[j,k] <- pow(pow(y_new[j,k],.5) - pow(eval[j,k],.5),2)

        # Chi-squared stat
        ch.err[j,k] <- pow((y[j,k] - eval[j,k]),2)/ (eval[j,k] + 0.5)
        ch.errnew[j,k] <- pow((y_new[j,k] - eval[j,k]),2)/ (eval[j,k]
        + 0.5)
    }
}

# Posterior predictive checks
Tobs <- sum(Terr[,])
Tnew <- sum(Terrnew[,])
Chisq.obs <- sum(ch.err[,])
Chisq.new <- sum(ch.errnew[,])

#############
# Derived parameters

# Sum within experimental sites for both treatments
for(t in 1:2){
    z.sum.S2[t] <- sum(z[1:12,t])
    z.sum.N2[t] <- sum(z[13:24,t])
}

# Calculate average across sites for each treatment
z.sum.C <- mean(c(z.sum.S2[2], z.sum.N2[1]))
z.sum.H <- mean(c(z.sum.S2[1], z.sum.N2[2]))

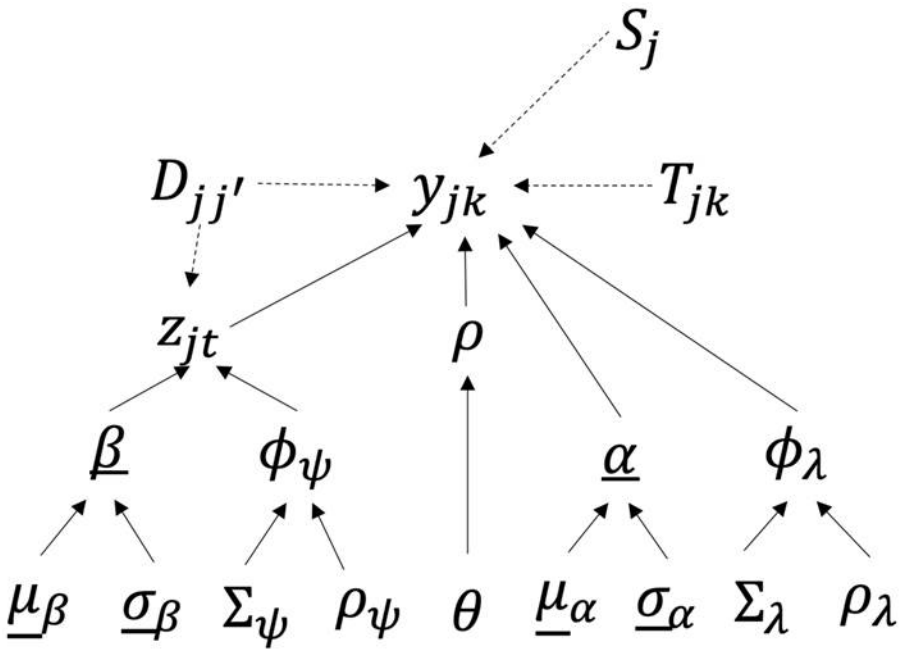
# Get average value of rho
rho.ave <- mean(rho[,])

# Estimate mean detection rate (lambda)
det.N2.c <- mean(lambda[13:24,1:5])
det.N2.h <- mean(lambda[13:24,6:10])
det.S2.h <- mean(lambda[1:12,1:5])
det.S2.c <- mean(lambda[1:12,6:10])

# Calculate average across sites for each treatment
```

```
    det.C <- mean(c(det.N2.c, det.S2.c))
    det.H <- mean(c(det.N2.h, det.S2.h))

    # Derived parameters for spatial correlation assessment
    eff.range.psi <- rho.psi * sqrt(-2 * log(0.05))
    eff.range.lambda <- rho.lambda * sqrt(-2 * log(0.05))
}
```



Directed acyclic graph of the zero-inflated negative binomial model used to evaluate elephant presence. Underscores denote vectors of random variables.